

Chapter 4 – Metasploit Framework

Part 1 – Introduction & Metasploit Fundamentals

What is Metasploit?

Metasploit Framework (MSF) is a penetration testing framework used to discover, validate, and assess vulnerabilities in authorized environments. It provides exploits, payloads, auxiliary modules, post-exploitation modules, encoders, listeners, and reporting capabilities.

Main Components

- Exploits: Target known vulnerabilities.
- Payloads: Code executed after successful exploitation.
- Auxiliary: Scanning and enumeration modules.
- Post: Actions after a session exists.
- Encoders: Modify payload representation (not a reliable AV bypass).
- NOP Generators: Internal support for some exploits/payloads.

Workflow

Reconnaissance → Scanning → Enumeration → Identify Vulnerability → Search Module → Configure → Execute (authorized) → Session → Post-Exploitation → Cleanup → Report.

Directory Structure

/usr/share/metasploit-framework/
modules/, exploits/, payloads/, auxiliary/, post/, encoders/, data/

Module Naming

Example: exploit/windows/smb/ms17_010_eternalblue
Type → Platform → Service → Vulnerability.

Database

Metasploit can use PostgreSQL to store hosts, services, credentials, vulnerabilities, notes, loot, and workspaces.

Normal Use

Use reconnaissance tools first, identify a vulnerability, configure the appropriate Metasploit module, verify exploitability in an authorized lab, perform post-exploitation if appropriate, and document findings.

Special Cases

If no module exists, a custom exploit may be required. Zero-day vulnerabilities generally require independent research. Manual exploitation may be preferable in some situations.

Common Beginner Mistakes

Skipping reconnaissance, choosing incompatible payloads, failing to configure required options, and assuming every target can be exploited.

Quick Revision

Metasploit is a framework—not just an exploit tool. Understand the vulnerability before using modules. Database features improve organization and reporting.